



Solving Equations with Complex Solutions

Some quadratic equations can be solved by finding the square root of each side of the equation.

Solving a Quadratic Equation with Real Solutions

The square root property says that if $x^2 = a$, then $x = \pm\sqrt{a}$. If $a > 0$, then the solutions for x are real numbers.

EXAMPLE 1

Solve $4x^2 = 80$ and check the solution.

SOLUTION

$$4x^2 = 80$$

$$x^2 = 20$$

Divide both sides of the equation by 4.

$$x = \pm\sqrt{20}$$

Take the square root of both sides.

$$x = \pm\sqrt{4} \cdot \sqrt{5}$$

Product Property of Square Roots

$$x = \pm 2\sqrt{5}$$

Simplify.

► **Remember** The product property of square roots states that

$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$$

for real numbers $a > 0$ and $b > 0$.

CHECK

Substitute $2\sqrt{5}$ for x .

$$\begin{aligned}4(2\sqrt{5})^2 &\stackrel{?}{=} 80 \\4 \cdot 20 &\stackrel{?}{=} 80 \\80 &= 80 \checkmark\end{aligned}$$

Substitute $-2\sqrt{5}$ for x .

$$\begin{aligned}4(-2\sqrt{5})^2 &\stackrel{?}{=} 80 \\4 \cdot 20 &\stackrel{?}{=} 80 \\80 &= 80 \checkmark\end{aligned}$$

The solution is correct. ■

Solving a Quadratic Equation with Complex Solutions

Quadratic equations such as $x^2 = -2$ have no real-number solutions because there is no square of a real number that equals -2 . Quadratic equations of this form have complex-number solutions.

Square Root Property for Imaginary Numbers

If $d > 0$, then $x^2 = -d$ has two imaginary solutions, $x = i\sqrt{d}$ and $x = -i\sqrt{d}$.

► **Think About It** When using the square root property, you can say you are “taking the square root of both sides.”

EXAMPLE 2

Solve $3x^2 + 21 = 0$ and check the solution.

SOLUTION

$$3x^2 + 21 = 0$$

$$3x^2 = -21 \quad \text{Subtract 21 from both sides.}$$

$$x^2 = -7 \quad \text{Divide both sides by 3.}$$

$$\sqrt{x^2} = \pm\sqrt{-7} \quad \text{Take the square root of both sides.}$$

$$x = \pm i\sqrt{7} \quad \text{Simplify.}$$

CHECK

Substitute $i\sqrt{7}$ for x . Substitute $-i\sqrt{7}$ for x .

$$3(i\sqrt{7})^2 + 21 \stackrel{?}{=} 0 \quad 3(-i\sqrt{7})^2 + 21 \stackrel{?}{=} 0$$

$$3(7i^2) + 21 \stackrel{?}{=} 0 \quad 3(7i^2) + 21 \stackrel{?}{=} 0$$

$$3(-7) + 21 \stackrel{?}{=} 0 \quad 3(-7) + 21 \stackrel{?}{=} 0$$

$$-21 + 21 \stackrel{?}{=} 0 \quad -21 + 21 \stackrel{?}{=} 0$$

$$0 = 0 \checkmark$$

$$0 = 0 \checkmark$$

The solution is correct. ■

EXAMPLE 3

Solve $(x + 3)^2 = -5$ and check the solution.

SOLUTION

$$(x + 3)^2 = -5$$

$$\sqrt{(x + 3)^2} = \pm\sqrt{-5} \quad \text{Take the square root of both sides.}$$

$$x + 3 = \pm i\sqrt{5} \quad \text{Simplify.}$$

$$x = -3 \pm i\sqrt{5} \quad \text{Subtract 3 from both sides.}$$

CHECK

Substitute $-3 + i\sqrt{5}$ for x . Substitute $-3 - i\sqrt{5}$ for x .

$$[(-3 + i\sqrt{5}) + 3]^2 \stackrel{?}{=} -5 \quad [(-3 - i\sqrt{5}) + 3]^2 \stackrel{?}{=} -5$$

$$(i\sqrt{5})^2 \stackrel{?}{=} -5 \quad (-i\sqrt{5})^2 \stackrel{?}{=} -5$$

$$5i^2 \stackrel{?}{=} -5 \quad 5i^2 \stackrel{?}{=} -5$$

$$5(-1) \stackrel{?}{=} -5 \quad 5(-1) \stackrel{?}{=} -5$$

$$-5 = -5 \checkmark$$

$$-5 = -5 \checkmark$$

The solution is correct. ■

► **Remember** The expressions $a + bi$ and $a - bi$ are a conjugate pair.

► **Think About It** Equations with complex solutions generally occur in conjugate pairs.

Finding an Equation, Given the Solutions

EXAMPLE 4

Find a polynomial equation in the form $p(z) = 0$ that has the given solutions.

A $z = \pm 8i$

SOLUTION

$$z = \pm 8i$$

$$z^2 = (\pm 8i)^2 \quad \text{Square both sides.}$$

$$z^2 = 64 \cdot i^2 \quad \text{Simplify.}$$

$$z^2 = -64$$

$$z^2 + 64 = 0 \quad \text{Put the equation into the right form.}$$

B $z = 3 \pm i\sqrt{7}$

SOLUTION

$$z = 3 \pm i\sqrt{7}$$

$$z - 3 = \pm i\sqrt{7} \quad \text{Subtract 3 from both sides.}$$

$$(z - 3)^2 = (\pm i\sqrt{7})^2 \quad \text{Square both sides.}$$

$$(z - 3)^2 = 7 \cdot i^2 \quad \text{Simplify.}$$

$$(z - 3)^2 = -7$$

$$z^2 - 6z + 16 = 0 \quad \text{Put the equation into the right form.} \quad \blacksquare$$